Learning Objectives:

At the end of this topic you will be able to;

☑ Generate the Boolean expression for a system [with up to 3 inputs] from a truth table;

☑ Generate the Boolean expression for a system [with up to 4 inputs] from a logic diagram;

☑ Recall and use the following identities

- $\mathbf{A.1 = A}$
- $\mathbf{A.0 = 0}$
- $\mathbf{A.\overline{A} = 0}$
- $\mathbf{A + 1 = 1}$
- $\mathbf{A + 0 = A}$
- $\mathbf{A + \overline{A} = 1}$

☑ Select and use the following identities

- $\mathbf{A + \overline{A}.B = A + B}$
- $\mathbf{A.B + A = A(B + 1) = A}$

☑ Apply DeMorgan's theorem to simplify a logic system [with up to 3 inputs]

Module ET1
Introduction to Analogue and Digital Systems.

Boolean Algebra.

In our previous unit we discovered that logic systems can become quite complex if we don't use any simplification techniques. In this topic you will be introduced to the first method of simplification, Boolean Algebra.

We have already introduced some of the ideas behind Boolean Algebra in simple terms in Topic 1.1.1, and 1.2.1. Here is a brief summary of the idea's we have covered so far.

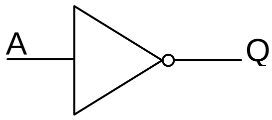
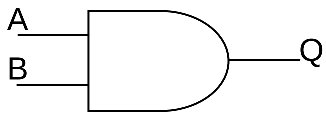
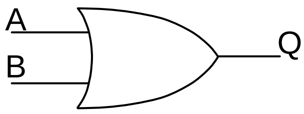
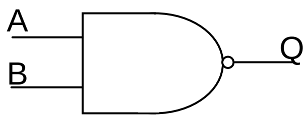
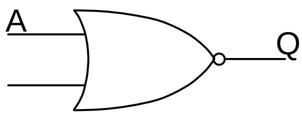
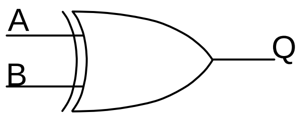
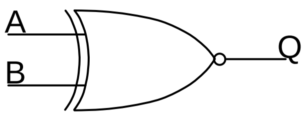
- A bar, on top of an input variable, is used to represent the NOT function. E.g. NOT **C** = $\overline{C}$
- A dot '.', is used to represent the AND function. E.g. **A** and **B** = $A.B$
- A plus '+', is used to represent the OR function. E.g. **E** OR **F** = $E+F$
- A plus with a circle '/', is used to represent the Exclusive OR (EXOR) function. E.g. **C** EXOR **D** = $C\oplus D$

Exercise 1: Write down the Boolean Expressions for the following examples:

1. NOT **X** ....
2. **R** AND **T** ....
3. **S** OR **R** ....
4. **C** EXOR **G** ....
5. NOT **B** AND **C** ....
6. NOT **X** OR NOT **Y** ....
7. **P** EXOR NOT **B** ....
8. NOT (**A** AND **B**) ....
9. **X** NOR **Y** ....

Topic 1.2.2 – System simplification using Boolean Algebra

The following table should remind you of the work we did in Topic 1.1.1.

Gate	Symbol	Boolean Equation
NOT		$\mathbf{Q} = \overline{\mathbf{A}}$
AND		$\mathbf{Q} = \mathbf{A.B}$
OR		$\mathbf{Q} = \mathbf{A} + \mathbf{B}$
NAND		$\mathbf{Q} = \overline{\mathbf{A.B}}$
NOR		$\mathbf{Q} = \overline{\mathbf{A} + \mathbf{B}}$
EXOR		$\mathbf{Q} = \mathbf{A} \oplus \mathbf{B}$ **or** $\mathbf{Q} = \overline{\mathbf{A}}.\mathbf{B} + \mathbf{A}.\overline{\mathbf{B}}$
EXNOR		$\mathbf{Q} = \overline{\mathbf{A} \oplus \mathbf{B}}$ **or** $\mathbf{Q} = \overline{\mathbf{A}}.\overline{\mathbf{B}} + \mathbf{A}.\mathbf{B}$

Now let us put this into practice. There are two ways in which Boolean expressions for a logic system can be formed, either from a truth table or from a logic circuit diagram. We will now consider each of these in turn starting with the easiest, which is to complete a Boolean expression from a truth table.

Deriving a Boolean Expression from a Truth Table

Consider the following truth table of a particular logic system. (Note : at this stage it does not matter what the system is meant to do).

C	B	A	Q
0	0	0	0
0	0	1	1
0	1	0	0
0	1	1	1
1	0	0	1
1	0	1	1
1	1	0	0
1	1	1	0

In this table there are four combinations of inputs that will produce an output at **Q**. In order to write down the Boolean Expression for the whole system we first have to write down the Boolean Equation for each line in the truth table where the output is a 1. All input variables (three in this case) must be included in each Boolean Equation on each line, as shown below

C	B	A	Q	Boolean Equation
0	0	0	0	
0	0	1	1	$\mathbf{A.\overline{B}.\overline{C}}$
0	1	0	0	
0	1	1	1	$\mathbf{A.B.\overline{C}}$
1	0	0	1	$\mathbf{\overline{A}.\overline{B}.C}$
1	0	1	1	$\mathbf{A.\overline{B}.C}$
1	1	0	0	
1	1	1	0	

To obtain the Boolean expression for the whole system we simply take each of these terms and 'OR' them together.

$$\mathbf{Q = A.\overline{B}.\overline{C} + A.B.\overline{C} + \overline{A}.\overline{B}.C + A.\overline{B}.C}$$

The expression we obtain from this may not be the simplest possible, we will look at simplification later. We will concentrate to start with on obtaining a correct Boolean expression for a logic system before we attempt to simplify them. After all there is not a lot of point being able to simplify an expression if this is not correct to start with.

Exercise 2: Here are a few for you to try:

1.

C	B	A	Q	Boolean Equation
0	0	0	1	
0	0	1	0	
0	1	0	0	
0	1	1	1	
1	0	0	0	
1	0	1	0	
1	1	0	1	
1	1	1	0	

Boolean Expression for system:

....

2.

C	B	A	Q	Boolean Equation
0	0	0	0	
0	0	1	0	
0	1	0	1	
0	1	1	0	
1	0	0	0	
1	0	1	1	
1	1	0	0	
1	1	1	1	

Boolean Expression for system:

....

3. Now try one with four inputs, the process is exactly the same, it's just that you now have four variables in each term.

D	C	B	A	Q	Boolean Equation
0	0	0	0	0	
0	0	0	1	0	
0	0	1	0	0	
0	0	1	1	0	
0	1	0	0	1	
0	1	0	1	0	
0	1	1	0	1	
0	1	1	1	0	
1	0	0	0	0	
1	0	0	1	0	
1	0	1	0	0	
1	0	1	1	1	
1	1	0	0	0	
1	1	0	1	0	
1	1	1	0	1	
1	1	1	1	0	

Boolean Expression for system:

……

Hopefully you will have found these tasks fairly straight forward as they are quite easy once you understand how to write down the variables in Boolean terms.

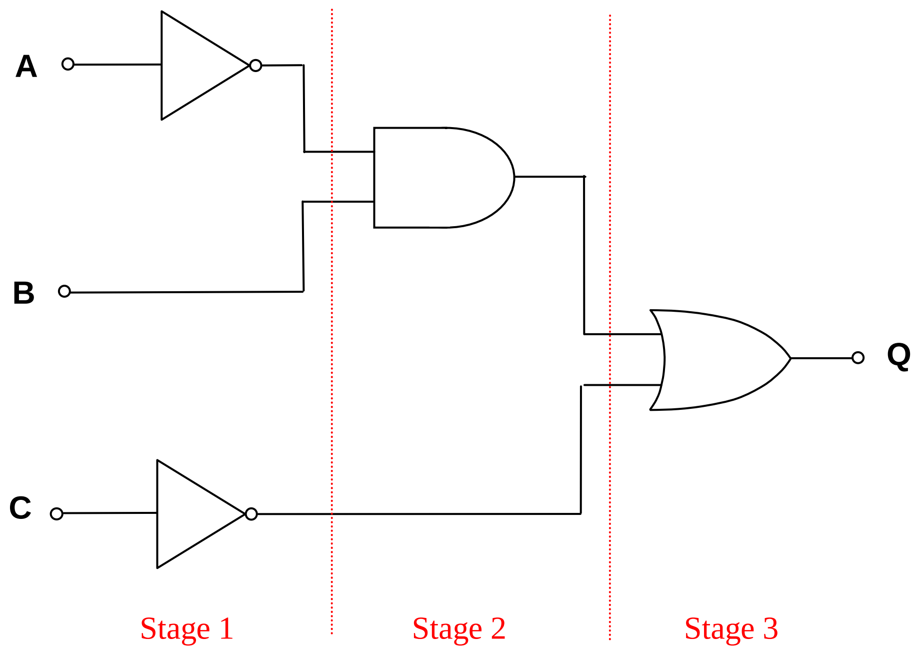
We will now look at how to generate the Boolean expressions from a logic circuit diagram, which can be a little more difficult depending on the type of gates involved.

Deriving a Boolean Expression from a Logic Circuit.

Sometimes we are not given a truth table, but a logic circuit diagram from which we have to derive the Boolean expression. This sounds complicated but as long as you are careful in what you do and work sensibly from the inputs across to the outputs you should be o.k.

Let us work through an example together.

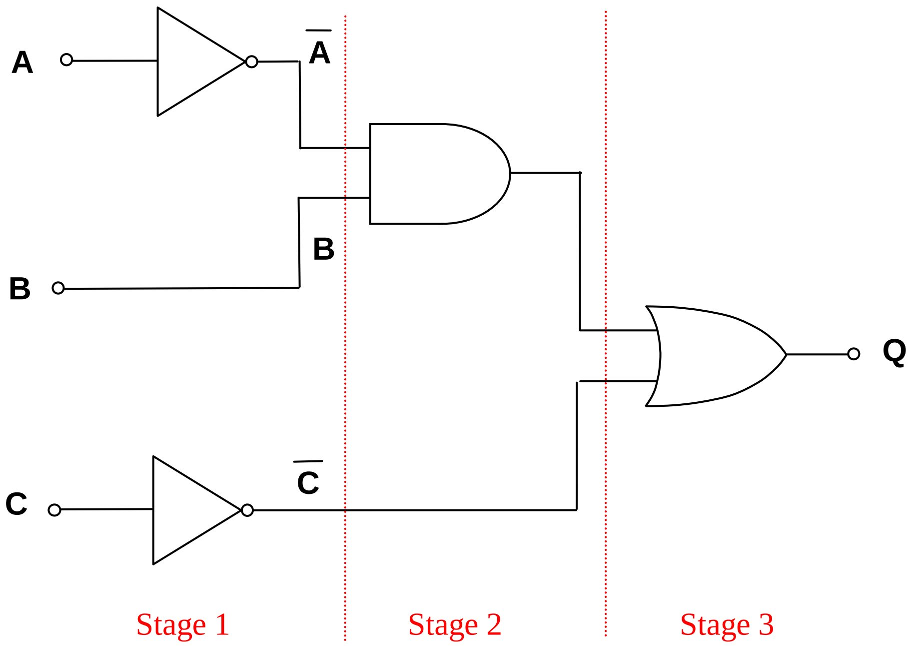
Example 1: Derive the Boolean expression for the following logic circuit.



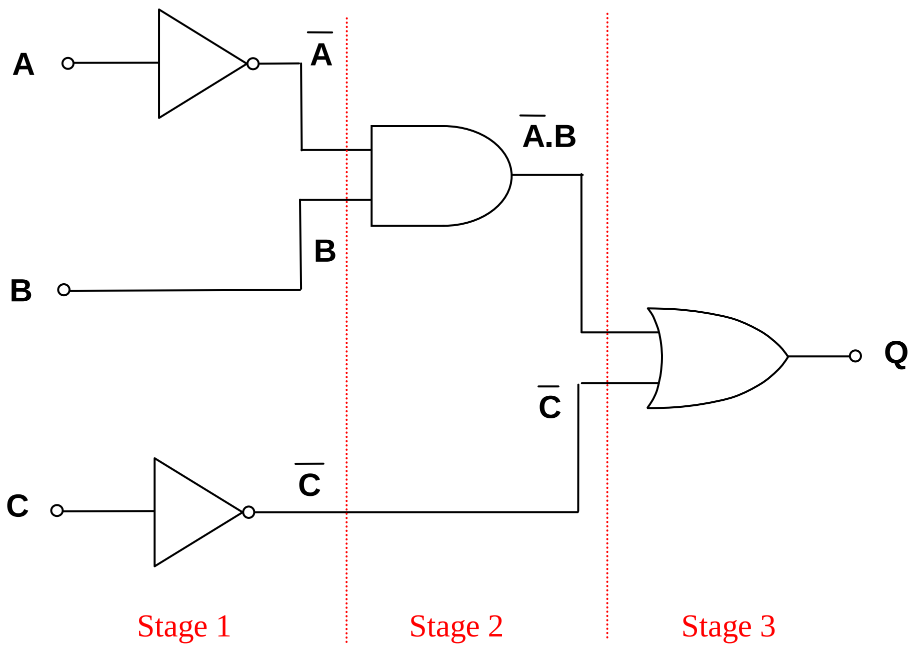
To start the process try to divide the circuit into stages as shown above.

This breaks down a large circuit into smaller more manageable chunks. It would be virtually impossible to write down the expression for Q immediately just by looking at the logic diagram, so don't even try, you are more than likely to make a mistake.

Having identified the different stages in the circuit diagram, we now proceed to write down the Boolean expressions for all of the output sections of stage 1, as shown on the following page.



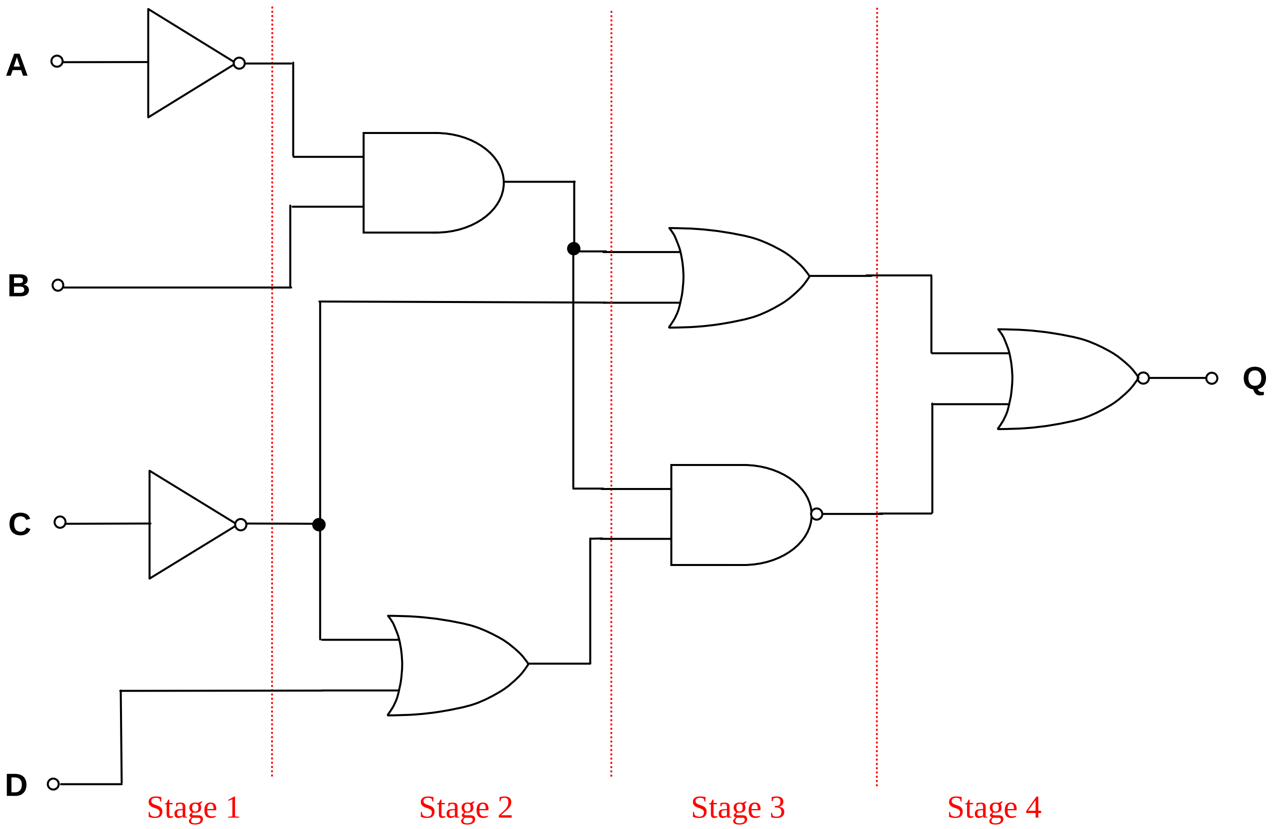
Now we move on to look at the outputs of stage 2, as shown below:



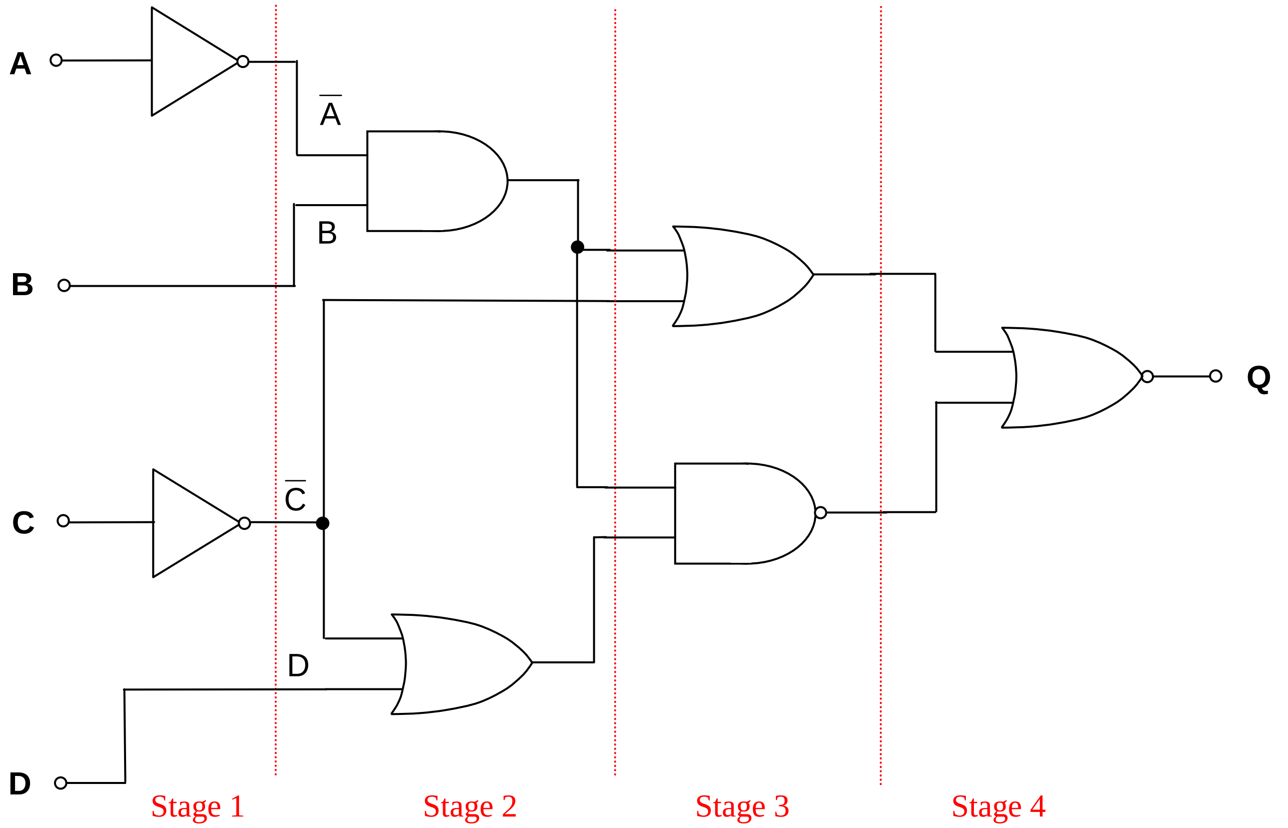
And finally we can complete stage 3, to arrive at the expression for the system as

$$Q = \overline{A}.B + \overline{C}$$

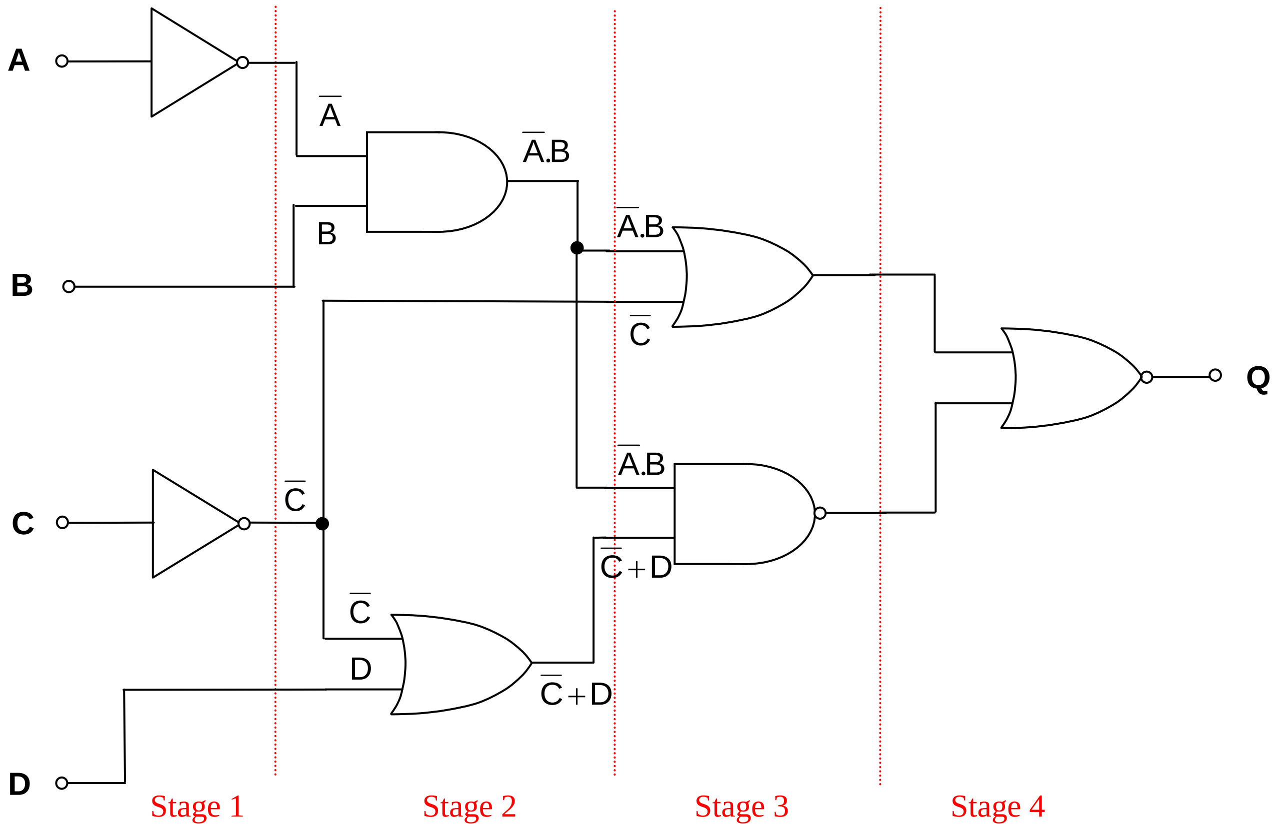
This was a relatively small system of logic gates, and the full system Boolean expression was determined quite quickly. Now consider a much larger system as shown below. Don't be put off, we will again work through this slowly, one stage at a time



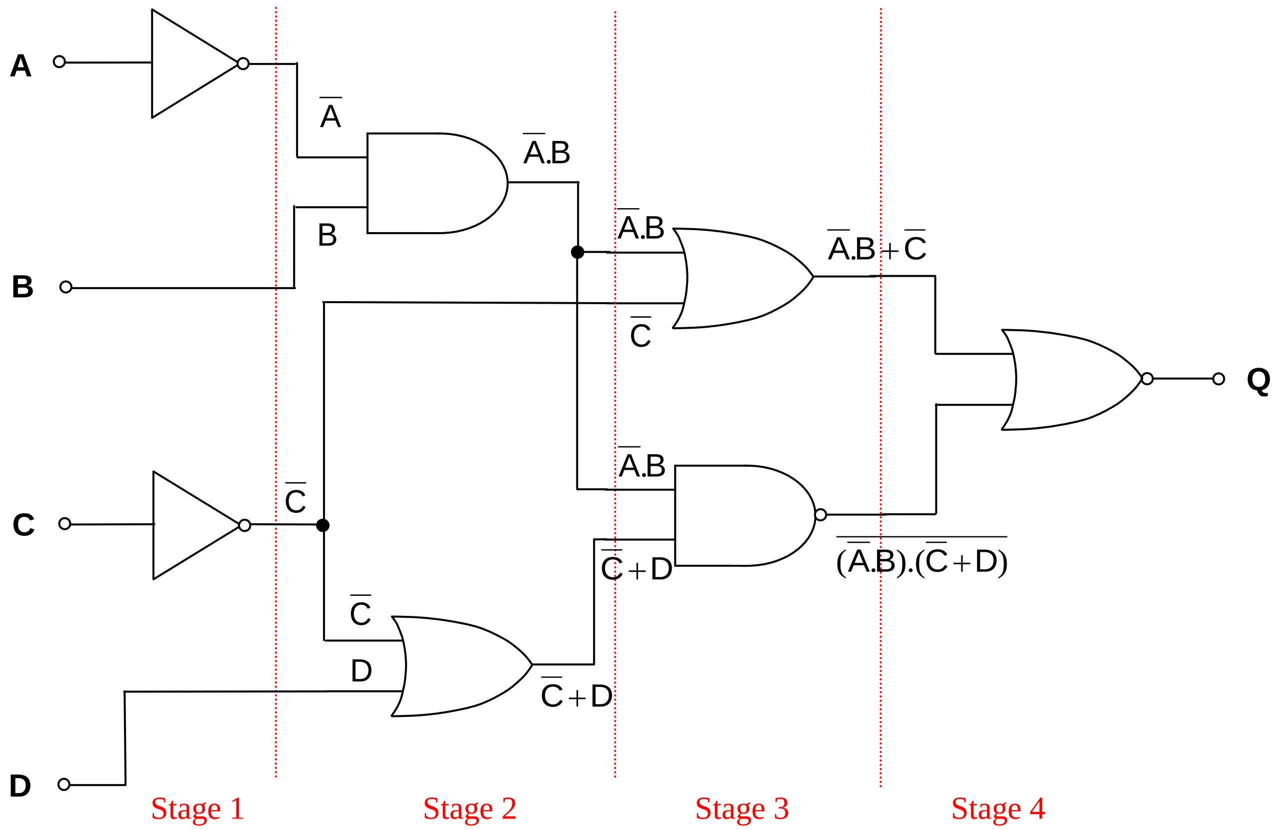
So to begin with we will work out all of the outputs in stage 1.



Then we look at stage 2.



Now for stage 3, we have to be very careful as for the first time we are going to come across NAND gates, and it is really important to keep terms together, so we use brackets to keep things together.



In stage 4 we combine these two large expressions with a NOR function. Again notice the use of brackets to keep terms together.

$$Q=\overline{(\overline{A}.B+\overline{C})+\overline{(\overline{A}.B).(\overline{C}+D)}}$$

There is no easy way of learning how to complete these logic expressions from circuit diagrams other than through practice. So the following examples should help you practice the skills needed.

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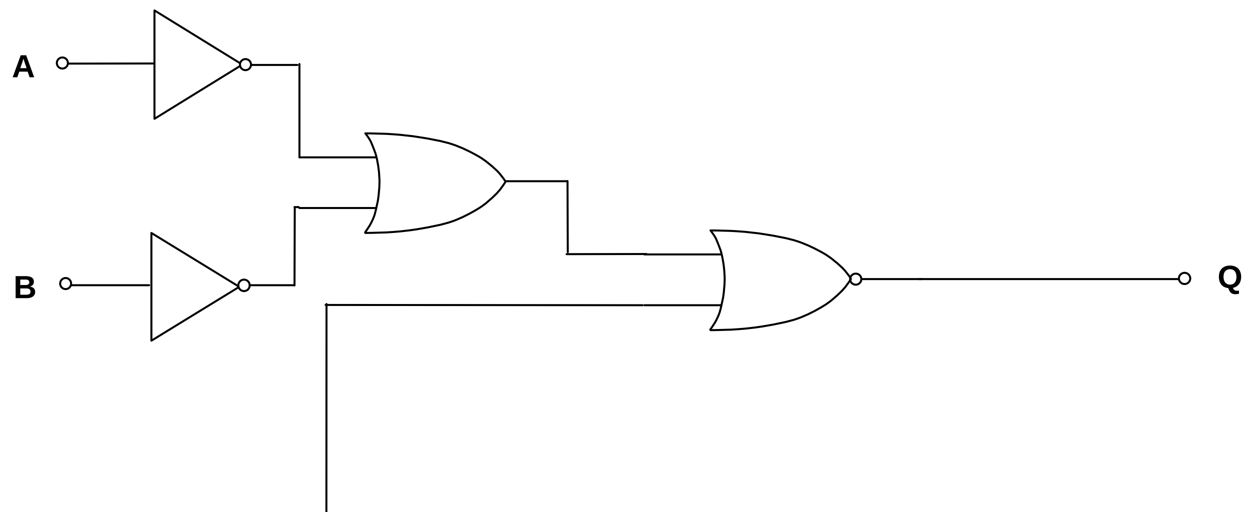
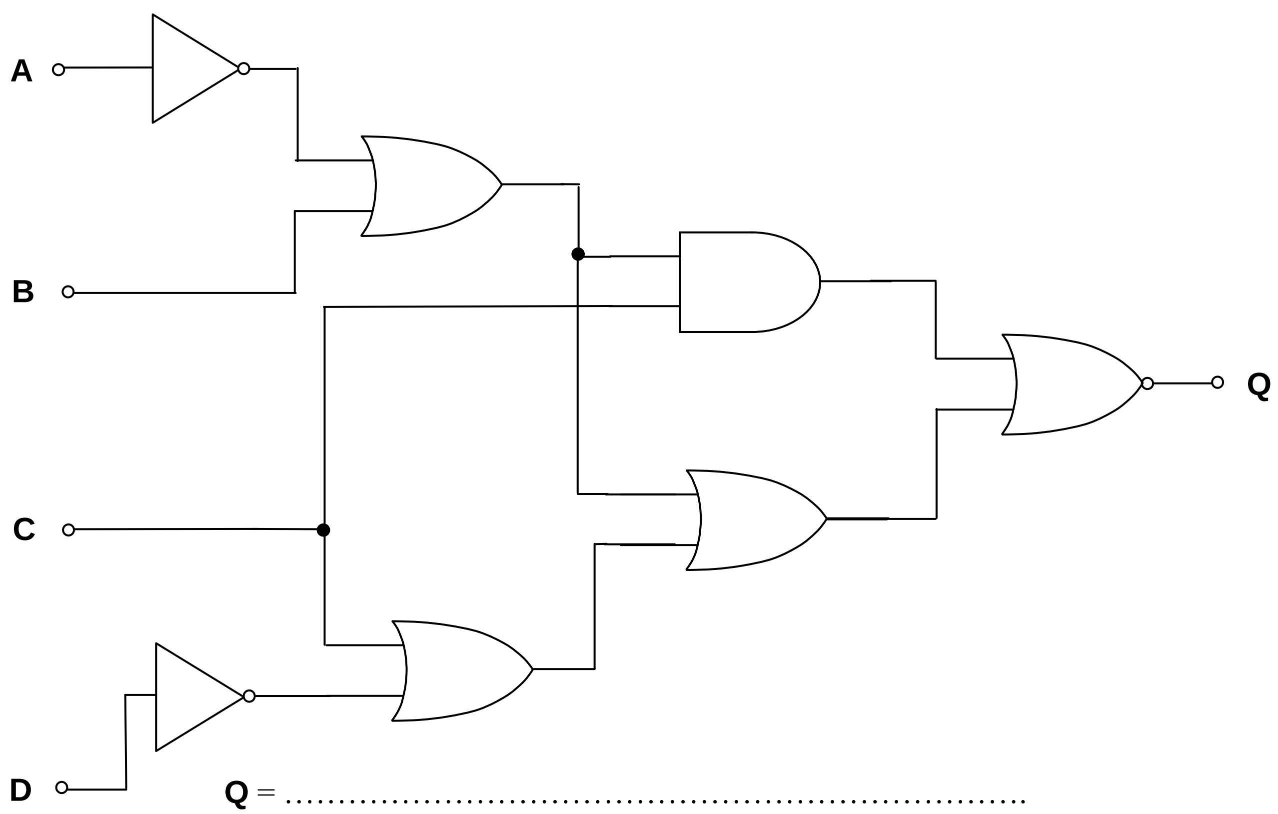
Exercise 3: Derive the Boolean Expression for the output of the following logic systems.

1.



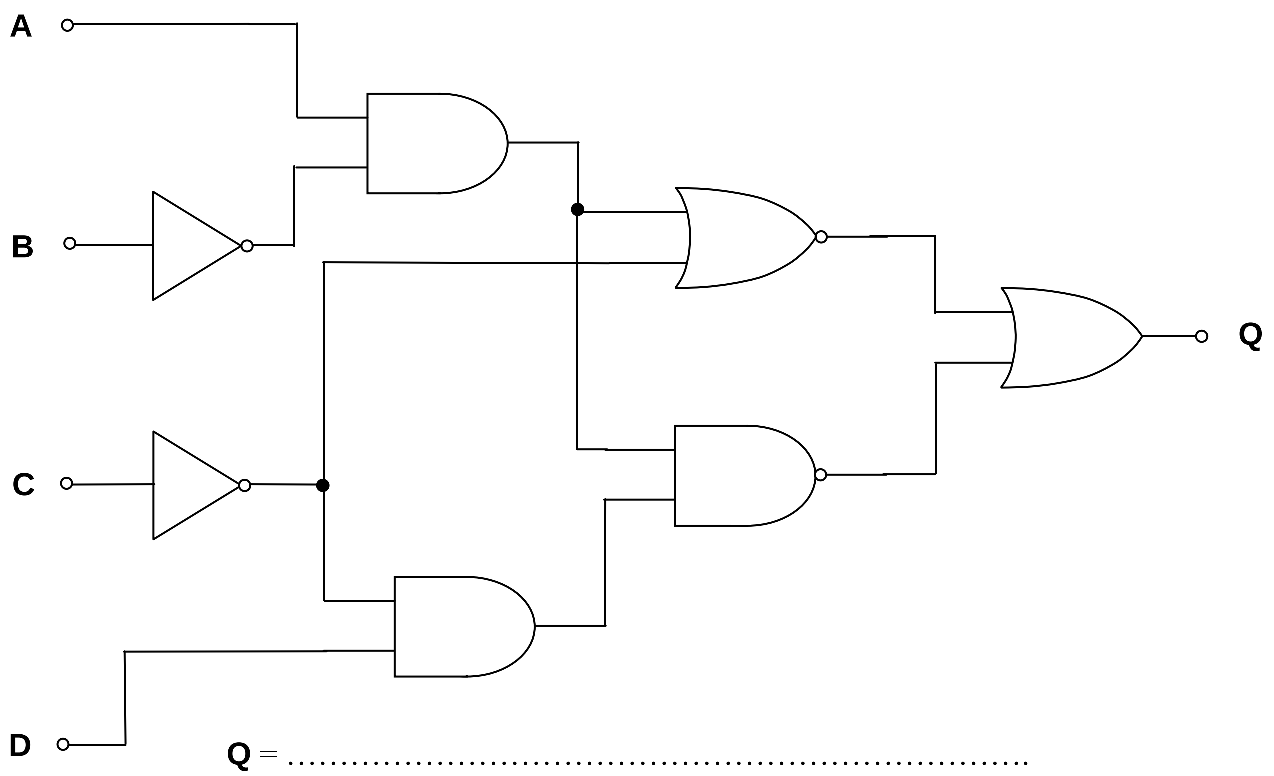
Q = ……………………………………………………………

2.



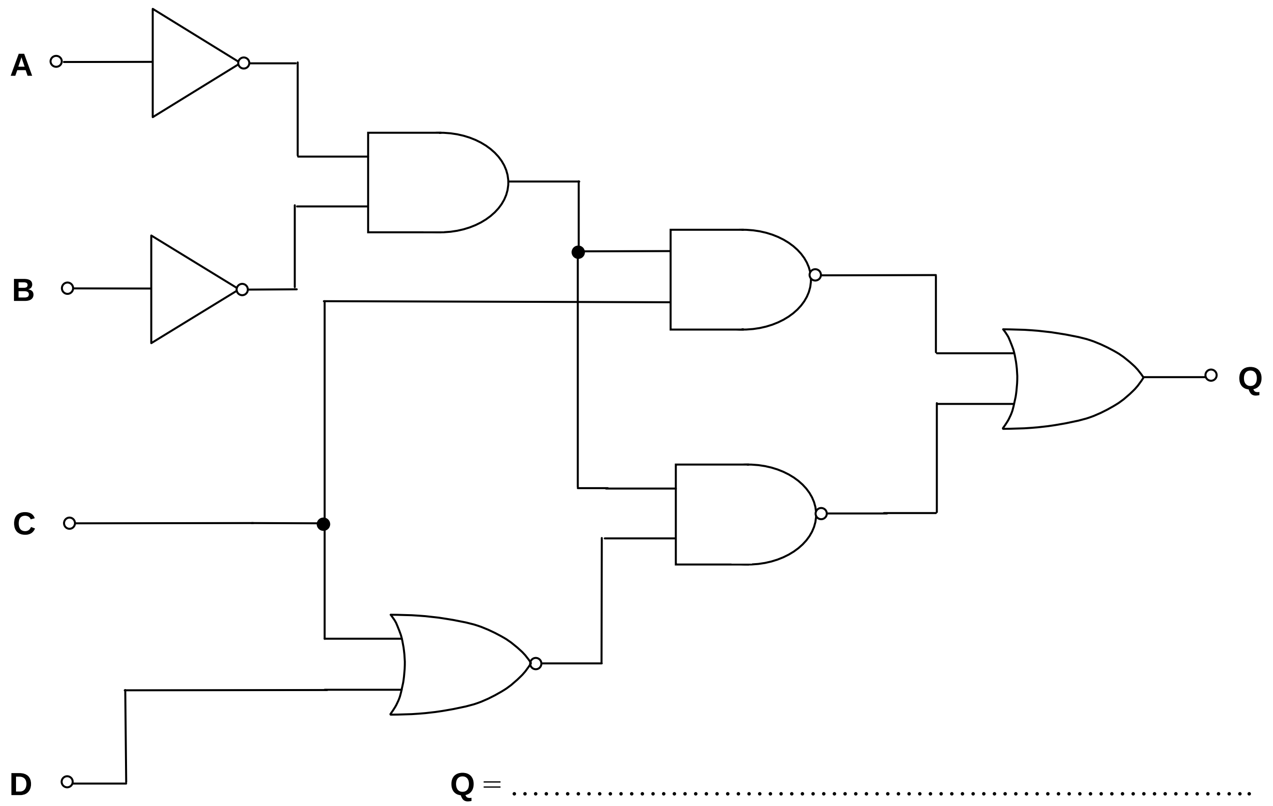
Q = ……………………………………………………………

3.



Q = ………………………………………………………………

4.



Q = ………………………………………………………………

So now we are able to derive a Boolean Expression for a logic system either from a truth table or from a logic diagram. The expression we end up with is often very long and complex and can result in very complicated logic systems being built. What is needed is a method of simplifying these logic expressions using Boolean algebra.

The introduction of the word algebra may remind many of you of the algebra you used in GCSE Mathematics to solve questions like:

simplify $2x + 6y$.

Giving an answer of $2(x+3y)$.

Or

$abc + bcd = bc(a+d)$

Or

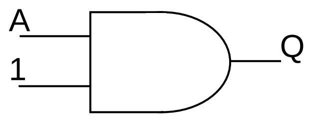
$2xy - 4yz = 2y(x+2z)$ etc.

The solution involves looking for common elements that can be removed from each term to simplify the expression by using brackets. The original expression can be regenerated by multiplying everything inside the bracket by the term outside. All we have done is to find an alternative way of writing down the expression. Boolean algebra is very similar, we try to remove common elements to simplify the expression. However there are a few things that we can do to enable the expression to become much simpler if we can remember some basic combinations that help to reduce terms significantly. These special terms are called **identities**, and you will have to remember them as they are not reproduced on the information sheet of the examination paper.

We will now have a look at these special identities.

Topic 1.2.2 – System simplification using Boolean Algebra

1. Consider the following Logic Circuit.



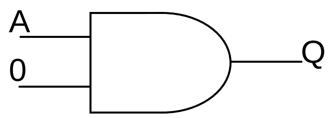
The output is **Q = A.1** in Boolean Algebra.

If **A** is a Logic 0, then **Q** will also be Logic 0, since 0.1 = 0,
If **A** is a Logic 1, then **Q** will also be a Logic 1, since 1.1 = 1.
The output **Q** will therefore be the same Logic Level as **A**.

So wherever the term **A.1** appears this can be replaced with **A** so our first identity is

A.1 = A

2. Consider the following Logic Circuit.



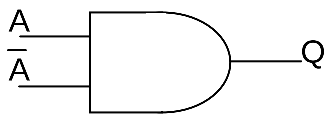
The output is **Q = A.0** in Boolean Algebra.

If **A** is a Logic 0, then **Q** will also be Logic 0, since 0.0 = 0,
If **A** is a Logic 1, then **Q** will be a Logic 0, since 1.0 = 0.
The output **Q** will therefore always be Logic 0

So wherever the term **A.0** appears this can be replaced with **0** so our second identity is

A.0 = 0

3. Consider the following Logic Circuit.



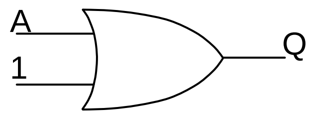
The output is $\mathbf{Q = A.\overline{A}}$ in Boolean Algebra.

If **A** is a Logic 0, then $\mathbf{\overline{A}}$ will be Logic 1, so **Q** will be Logic 0, since 0.1 = 0,
If **A** is a Logic 1, then $\mathbf{\overline{A}}$ will be Logic 0, so **Q** will be Logic 0, since 1.0 = 0,
The output **Q** will therefore always be Logic 0

So wherever the term $\mathbf{A.\overline{A}}$ appears this can be replaced with **0** so our third identity is

$$\mathbf{A.\overline{A} = 0}$$

4. Consider the following Logic Circuit.



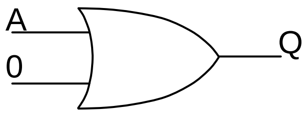
The output is **Q = A+1** in Boolean Algebra.

If **A** is a Logic 0, then **Q** will also be Logic 1, since 0+1 = 1,
If **A** is a Logic 1, then **Q** will also be Logic 1, since 1+1 = 1.
The output **Q** will therefore always be Logic 1

So wherever the term **A+1** appears this can be replaced with **1** so our fourth identity is

$$\mathbf{A+1 = 1}$$

5. Consider the following Logic Circuit.



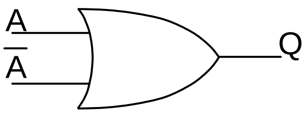
The output is **Q = A+0** in Boolean Algebra.

If **A** is a Logic 0, then **Q** will also be Logic 0, since 0+0 = 0,
If **A** is a Logic 1, then **Q** will also be Logic 1, since 0+1 = 1.
The output **Q** will therefore be the same Logic Level as **A**.

So wherever the term **A+0** appears this can be replaced with **A** so our fifth identity is

$$\mathbf{A+0 = A}$$

6. Consider the following Logic Circuit.



The output is $\mathbf{Q = A + \overline{A}}$ in Boolean Algebra.

If **A** is a Logic 0, then $\mathbf{\overline{A}}$ will be Logic 1, so **Q** will be Logic 1, since 0+1 = 1,
If **A** is a Logic 1, then $\mathbf{\overline{A}}$ will be Logic 0, so **Q** will be Logic 1, since 1+0 = 1,
The output **Q** will therefore always be Logic 1.

So wherever the term $\mathbf{A + \overline{A}}$ appears this can be replaced with **1** so our sixth identity is

$$\mathbf{A + \overline{A} = 1}$$

In all of these examples the variable A has been used, but the rules apply for any variable. i.e. $\mathbf{B.0 = 0}$, $\mathbf{C + 1 = 1}$, $\mathbf{\overline{D} + 0 = \overline{D}}$, $\mathbf{E.\overline{E} = 0}$ etc.
Now let's use these to simplify some Boolean expressions.

Examples :

1. Simplify the following expression.

$$Q = A.B.\overline{C} + A.B.C + A.\overline{B}$$

Simplification can begin by looking for common terms, in this case *A* is common to all terms so we can remove this outside a bracket as shown below:

$$Q = A.(B.\overline{C} + B.C + \overline{B})$$

Now we can see that B is common to the first two terms inside the bracket, but it is not common to the last term so we cannot include this in the simplification, which now becomes:

$$Q = A.(B.(\overline{C} + C)) + \overline{B})$$

Using our sixth identity the term $\overline{C} + C = 1$ so the expression now becomes:

$$Q = A.(B.(1) + \overline{B})$$

Using the first identity **B.1 = B** so the expression becomes

$$Q = A.(B + \overline{B})$$

Using our sixth identity again the term $B + \overline{B} = 1$ so the expression now becomes:

$$Q = A.1$$

Using the first identity **A.1 = A** so the expression finally becomes

$$Q = A$$

Using this mathematical approach the solution where common factors were identified in all terms actually made the expression a little bit more complex with the insertion of all the brackets, before we were able to start simplification. If we group pairs of terms instead of the whole, with intention of obtaining identities such as $A + 1$or $A + \overline{A}$ inside the brackets we often reach the simplest solution, without having to insert multiple brackets as shown below. We will solve the same example by this different technique.

Simplify the following expression.

$$\mathbf{Q = A.B.\overline{C} + A.B.C + A.\overline{B}}$$

Simplification can begin by combining the first two terms only as follows:

$$\mathbf{Q = A.B.(\overline{C} + C) + A.\overline{B}}$$

Using our sixth identity the term $\mathbf{C + \overline{C} = 1}$ so the expression now becomes:

$$\mathbf{Q = A.B.1 + A.\overline{B}}$$

Using the first identity **A.B.1 = A.B** so the expression becomes

$$\mathbf{Q = A.B + A.\overline{B}}$$

Now we can remove the common term again to leave:

$$\mathbf{Q = A.(B + \overline{B})}$$

Using our sixth identity the term $\mathbf{B + \overline{B} = 1}$ so the expression now becomes:

$$\mathbf{Q = A.1}$$

Using the first identity **A.1 = A** so the expression finally becomes

$$\mathbf{Q = A}$$

I'm sure you'll agree this is a lot simpler than $\mathbf{Q = A.B.\overline{C} + A.B.C + A.\overline{B}}$. Hopefully you will see the advantage of simplifying these expressions whenever we have the opportunity. It is normal practice in ***electronics*** to take the smaller groups and remove common factors to leave a useful combination behind like $\mathbf{B + \overline{B} = 1}$, as this means the combination can be removed. Let's try another one!

2. Simplify the following expression:

$$Q = \overline{A}.B + A.B.\overline{C} + A.B + C$$

Start by looking for possible common factors which will leave behind one of our two key identities inside the bracket, in this case **A.B** in terms 2 and 3. This gives the following simplification.

$$Q = \overline{A}.B + A.B.(\overline{C} + 1) + C$$

Using our fourth identity the term $\overline{C} + 1 = 1$ so the expression now becomes:

$$Q = \overline{A}.B + A.B.1 + C$$

Using the first identity **A.B.1 = A.B** so the expression becomes

$$Q = \overline{A}.B + A.B + C$$

Again we look for common terms, this time between the first two terms to give the following:

$$Q = B.(\overline{A} + A) + C$$

Using our sixth identity the term $A + \overline{A} = 1$ so the expression now becomes:

$$Q = B.1 + C$$

Using the first identity **B.1 = B** so the expression becomes

$$Q = B + C$$

And so we arrive at the simplest solution once again. You would not be expected to write out all of the steps as has been shown here in your own simplifications. These have only been included to show you how each step has been arrived at.

In the next two examples these intervening descriptions will be removed, to show you what typical solutions look like.

3. Simplify the following expression.

$$Q = \overline{A}.B + \overline{A}.B.C + \overline{A}.\overline{B}.\overline{C} + A.\overline{B}.\overline{C}$$

Solution:

$$Q = \overline{A}.B + \overline{A}.B.C + \overline{A}.\overline{B}.\overline{C} + A.\overline{B}.\overline{C}$$
$$Q = \overline{A}.B + \overline{A}.B.C + \overline{B}.\overline{C}.(\overline{A} + A)$$
$$Q = \overline{A}.B + \overline{A}.B.C + \overline{B}.\overline{C}.1$$
$$Q = \overline{A}.B.(1 + C) + \overline{B}.\overline{C}$$
$$Q = \overline{A}.B.1 + \overline{B}.\overline{C}$$
$$Q = \overline{A}.B + \overline{B}.\overline{C}$$

4. Simplify the following expression.

$$Q = B.C.(\overline{C} + D) + C.D + C + \overline{A}$$

Solution:

$$Q = B.C.(\overline{C} + D) + C.D + C + \overline{A}$$
$$Q = B.C.\overline{C} + B.C.D + C.D + C + \overline{A}$$
$$Q = B.0 + B.C.D + C.(D + 1) + \overline{A}$$
$$Q = B.C.D + C + \overline{A}$$
$$Q = C.(B.D + 1) + \overline{A}$$
$$Q = C.1 + \overline{A}$$
$$Q = C + \overline{A}$$

In this last example, notice that the original expression first needed to be expanded before the expression could be simplified. This is sometimes necessary to be able to group terms together in different ways.

The process may at first seem complicated and you may find it hard in the beginning to identify the terms to group together, but this comes from experience and the old phrase 'practice makes perfect' is very true here. So why don't you try the next few examples.

Exercise 4 : Simplify the following Boolean expressions.

1. $Q = A.B + B$

....

....

2. $Q = C.(A + \overline{C})$

....

....

....

3. $Q = A.B.\overline{C} + A.\overline{C} + \overline{A}.\overline{C}.D + \overline{A}.\overline{C}.\overline{D}$

....

....

....

....

....

....

4. $Q = A.B.(\overline{B} + C) + B.C + B$

....

....

....

....

....

....

5. $\mathbf{Q = B.(A + \overline{C}) + A + A.(\overline{A} + B)}$

6. $\mathbf{Q = A.B.\overline{C} + B.C + \overline{A}.B.\overline{C} + A.B.\overline{B}}$

7. $\mathbf{Q = A.B.C + B.C.D + B.C.\overline{D} + B.\overline{C}.D + A.B.\overline{C} + \overline{A}.B.\overline{C}}$

8. $\mathbf{Q = A.B.C.D + A.B.D + A.\overline{B}.D + A.\overline{B}.C.D + A.C.D + \overline{A}.C.D}$

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We have now looked at a number of examples of simplification using Boolean Algebra. The last couple of examples were quite complex, and it took a lot of processing to reach the final simplified expression. It would be nice if we could make this process easier, and the good news is that there is an easier method we can use which will be covered in our next Topic 1.2.3. However, you still need to know how to use the rules of Boolean Algebra as there will be questions on this in the examination, so I'm afraid you can't ignore it.

Two Special Identities.

There are two special identities that we have not discussed in detail yet, even though we have used one of them already, in our simplification process. These are special because they will be printed on the information page on the exam paper so you don't have to remember them. The first of these is as follows:

$$\mathbf{A.B + A = A(B + 1) = A}$$

This is quite a straightforward example but because more than one term is involved, it has been decided to provide this for candidates just in case they forget what to do in this case.

The second special identity is a little more difficult to explain. Let us take a look at the identity first.

$$\mathbf{A + \overline{A}.B = A + B}$$

This simplification can be explained if we consider the condition of variable **A**.

If **A** is a Logic 0, then the expression becomes **0 + 1.B = B**

If **A** is a Logic 1, then the expression becomes **1 + 0.B = 1**, i.e. the same as **A**.

The $\overline{\mathbf{A}}$ term is redundant in the original expression, and can therefore be deleted, since it cannot influence the output in any way.
This identity can appear in several different forms, for example:

$$\mathbf{\overline{A} + A.B = \overline{A} + B}$$
$$\mathbf{B.\overline{C} + C = B + C}$$
$$\mathbf{\overline{A}.\overline{C} + A = \overline{C} + A}$$

In every case you will notice that one variable appears in both terms and it is inverted in one of these terms. The simplification requires the removal of this variable from the AND term.

We can further prove this by considering the truth table for the same function as shown below:

Consider the function : $Q = A.\overline{B} + B = A + B$

A	B	$\overline{B}$	$A.\overline{B}$	$A.\overline{B}+B$	$A+B$
0	0	1	0	0	0
0	1	0	0	1	1
1	0	1	1	1	1
1	1	0	0	1	1

If we look at the results in the last two columns we can see that they are identical. This means that the expression $A.\overline{B} + B$ must be equivalent to $A + B$. This identity is easily forgotten but it is surprising how many times we need to use it when simplifying expressions, which is why it has been identified as being special, and printed on the information sheet of the exam paper for you to use. Here is a quick example to show you how this can be helpful.

Simplify the expression : $Q = A.B.C + A.\overline{C} + C.(D + \overline{C}) + A$

Solution.

$$Q = A.B.C + A.\overline{C} + C.(D + \overline{C}) + A$$
$$Q = A.(B.C + \overline{C}) + C.D + C.\overline{C} + A$$
$$Q = A.(B + \overline{C}) + C.D + 0 + A$$
$$Q = A.B + A.\overline{C} + C.D + A$$
$$Q = A.(B + 1) + A.\overline{C} + C.D$$
$$Q = A + A.\overline{C} + C.D$$
$$Q = A.(1 + \overline{C}) + C.D$$
$$Q = A + C.D$$

Topic 1.2.2 – System simplification using Boolean Algebra

In all of the examples we have considered so far we have only dealt with AND, OR and NOT gates, however, as we know there are also NAND and NOR gates that can be used in logic circuits. When these inverted gates are used the inversion takes place after the AND or OR operation has been completed which results in logic expressions like those shown below.

$$Q = \overline{A.B}$$

and

$$Q = \overline{C + D}$$

When these terms appear in a logic expression, it is not possible to remove one of the terms from under the 'bar' and leave the other behind. For example:

$$\overline{A.B} + \overline{A} \neq \overline{A}.(\overline{B} + 1) = \overline{A}$$

Completion of the truth table below shows this to be the case.

A	B	$\overline{A}$	$\overline{A.B}$	$\overline{A.B} + \overline{A}$
0	0	1	1	1
0	1	1	1	1
1	0	0	1	1
1	1	0	0	0

Clearly the two highlighted columns are not the same, and therefore we cannot deal with these terms in the same way. Does this mean that we cannot simplify expressions that have NAND or NOR functions in them ?

It would be fairly pointless if we couldn't since many logic circuits contain NAND and NOR gates, however we need to use another rule, in order to achieve our aim of simplifying circuits involving these gates.

DeMorgan's Theorem

DeMorgan's theorem allows the logic functions of NAND and NOR to be simplified. We do not have to go into in depth analysis of how DeMorgan arrived at his theorem, just be able to use the result which is surprisingly simple. The rules are as follows:

i. If you break a 'bar' change the sign underneath the break.

ii. If you complete a 'bar' change the sign underneath where the bar is joined. (We will use this in Topic 1.2.4)

Let's look at a couple of simple examples.

1. If we start with $\overline{A.B}$, DeMorgan's theorem suggests that this can be written as $\overline{A}+\overline{B}$. The 'bar' has been broken and the sign changed underneath the break in the bar. We can check this by looking at the truth table below.

A	B	$\overline{A}$	$\overline{B}$	$\overline{A.B}$	$\overline{A}+\overline{B}$
0	0	1	1	1	1
0	1	1	0	1	1
1	0	0	1	1	1
1	1	0	0	0	0

This shows that the two logic expressions are the same.

2. If we start with $\overline{A+B}$, DeMorgan's theorem suggests that this can be written as $\overline{A}.\overline{B}$. The 'bar' has been broken and the sign changed underneath the break in the bar. We can check this by looking at the truth table below.

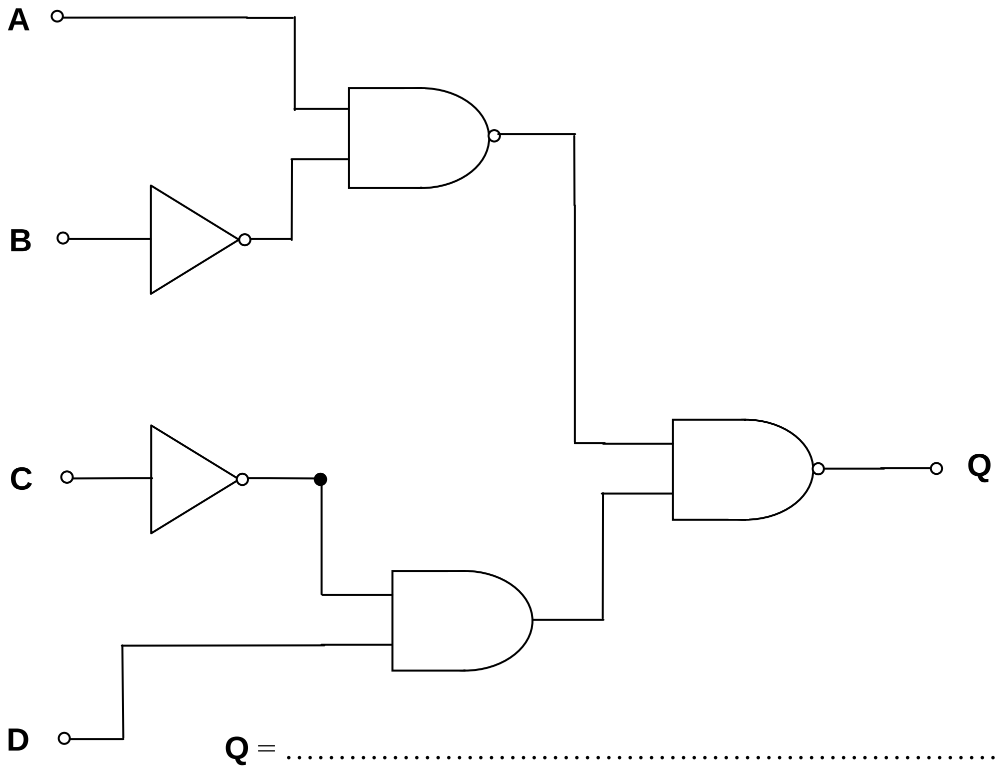
A	B	$\overline{A}$	$\overline{B}$	$\overline{A+B}$	$\overline{A}.\overline{B}$
0	0	1	1	1	1
0	1	1	0	0	0
1	0	0	1	0	0
1	1	0	0	0	0

This shows that the two logic expressions are the same.

The rule seems to work, according to the truth tables but we have only used very basic logic expressions here to prove the rule. If we looked at a more realistic problem then this would involve more terms and a more complex expression.

We will put a few of our new skills to the test by looking at a typical problem. You can work out some of the parts as we go through the example, but the correct solution will be provided at each stage so that you can check that you are on the correct path.

Consider the following logic circuit.



Q = ……………………………………………………………………

Use the process we looked at earlier to derive a Boolean expression for the logic circuit shown above.

Remember to be very careful with the NAND gates.

The Boolean expression you should have arrived at is as follows:

$$\mathbf{Q} = \overline{(\overline{\mathbf{A}.\overline{\mathbf{B}}}).(\overline{\mathbf{C}}.\mathbf{D})}$$

This looks quite complicated but by using DeMorgan's theorem we can make this a little bit easier. As a general 'rule of thumb' start from the top and work downwards. So the first thing that we have to do is to break to top 'bar' between the two brackets. This gives us the following:

$$\mathbf{Q} = (\overline{\overline{\mathbf{A}.\overline{\mathbf{B}}}}) + (\overline{\overline{\mathbf{C}}.\mathbf{D}})$$

If we look at the first term we can see that there is a double 'bar' over the expression. This means that the term is inverted and then inverted again, which will return the original state of the term. Therefore the **double inversion** as it is called can be removed.

$$\mathbf{Q} = (\mathbf{A}.\overline{\mathbf{B}}) + (\overline{\overline{\mathbf{C}}.\mathbf{D}})$$

We now apply DeMorgan's theorem again to the second term to give the following:

$$\mathbf{Q} = (\mathbf{A}.\overline{\mathbf{B}}) + (\overline{\overline{\mathbf{C}}} + \overline{\mathbf{D}})$$

Again we have a double inversion, applied only to the variable **C**, which can be removed to leave the final expression as:

$$\mathbf{Q} = (\mathbf{A}.\overline{\mathbf{B}}) + (\mathbf{C} + \overline{\mathbf{D}})$$
$$\mathbf{Q} = \mathbf{A}.\overline{\mathbf{B}} + \mathbf{C} + \overline{\mathbf{D}}$$

If there were additional terms in the expression, this could now be simplified using the normal rules of Boolean algebra as discussed previously.

Now for a couple of examples for you to try!

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Module ET1
Introduction to Analogue and Digital Systems.

Exercise 5. – Simplify the following expressions as much as possible.

1. $\mathbf{Q} = \overline{(\overline{\mathbf{A}+\overline{\mathbf{B}}}).(\overline{\mathbf{A}.\overline{\mathbf{B}}})} + \overline{\mathbf{A}}.\mathbf{B}$

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2. $\mathbf{Q} = \overline{\overline{\overline{\mathbf{A}.\overline{\mathbf{C}}}.\overline{\mathbf{B}}}.\mathbf{D}} + \overline{\overline{\mathbf{C}}.\mathbf{D}}$

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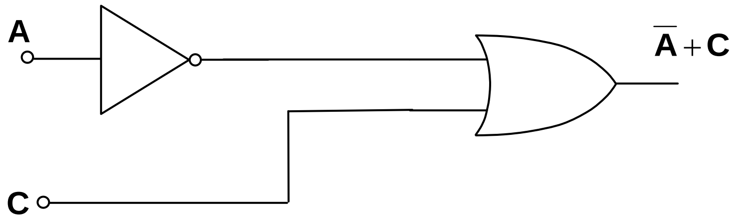
Creating a Logic Circuit Diagram From Boolean Equation.

Once the simplest Boolean expression has been obtained from a logic description or a truth table, you will often need to draw the logic circuit diagram. This just means working backwards from the Boolean expression, and as a general rule of thumb we start with any bracketed terms, then AND and finally OR functions. Let's have a look at a couple of examples.

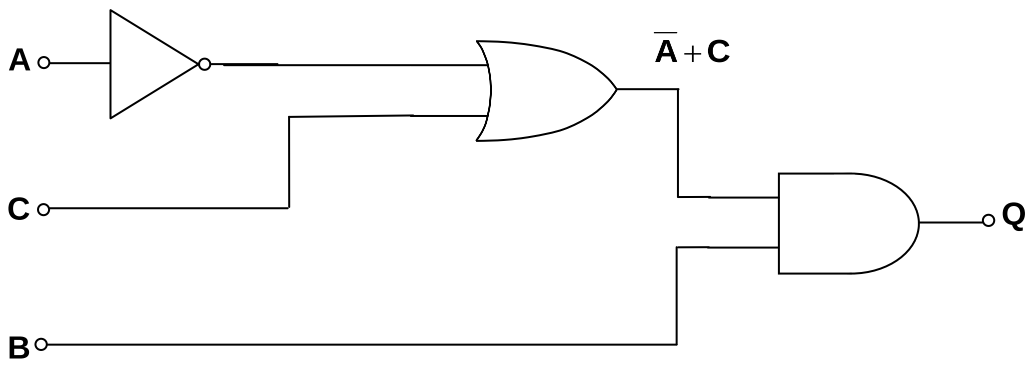
Example 1: Draw the logic diagram for the simplified Boolean expression shown below:

$$Q = B.(\overline{A} + C)$$

First we make up the logic for the bracketed term.



The output of this arrangement then needs to be connected to an AND gate with input **B** as shown below:



Here are a couple for you to try!

Exercise 6: Draw the Logic Circuit diagram for the Boolean expressions given.

1. $\mathbf{Q = A.\overline{B} + B.C}$

2. $\mathbf{Q = \overline{A + B} + \overline{A}.(C + B)}$

We have now covered all of the material needed for system simplification via Boolean Algebra. Together with the work we have done on logic system design, we are now in a position to tackle some of the examination style questions that brings all of the work we have done so far together.

The solutions which follow are the answers to the exercises given in this topic, and after those will be a series of examination style questions for you to practice with.

Module ET1
Introduction to Analogue and Digital Systems.

Solutions to Exercises:

Exercise 1.

1. $\overline{X}$
2. $R.T$
3. $S+R$
4. $C \oplus G$
5. $\overline{B}.C$
6. $\overline{X}+\overline{Y}$
7. $P \oplus \overline{B}$
8. $\overline{(A.B)}$
9. $\overline{X+Y}$

Exercise 2.

1. $Q=\overline{A}.\overline{B}.\overline{C}+A.B.\overline{C}+\overline{A}.B.C$
2. $Q=\overline{A}.B.\overline{C}+A.\overline{B}.C+A.B.C$
3. $Q=\overline{A}.\overline{B}.C.\overline{D}+\overline{A}.B.C.\overline{D}+A.B.\overline{C}.D+\overline{A}.B.C.D$

Exercise 3.

1. $Q=\overline{C+(\overline{A}+\overline{B})}$
2. $Q=\overline{(\overline{A}+B).C+(\overline{A}+B)+(C+\overline{D})}$
3. $Q=\overline{A.\overline{B}+\overline{C}}+\overline{(A.\overline{B}).(D.\overline{C})}$
4. $Q=\overline{(\overline{A}.\overline{B}).C}+\overline{(\overline{A}.\overline{B}).(\overline{C+D})}$

Exercise 4.

1.

$$Q=A.B+B$$
$$Q=B.(A+1)$$
$$Q=B.1$$
$$Q=B$$

2.

$$Q = C.(A + \overline{C})$$
$$Q = A.C + C.\overline{C}$$
$$Q = A.C + 0$$
$$Q = A.C$$

3.

$$Q = A.B.\overline{C} + A.\overline{C} + \overline{A}.\overline{C}.D + \overline{A}.\overline{C}.\overline{D}$$
$$Q = A.\overline{C}.(B + 1) + \overline{A}.\overline{C}.(D + \overline{D})$$
$$Q = A.\overline{C}.1 + \overline{A}.\overline{C}.1$$
$$Q = A.\overline{C} + \overline{A}.\overline{C}$$
$$Q = \overline{C}.(A + \overline{A})$$
$$Q = \overline{C}.1$$
$$Q = \overline{C}$$

4.

$$Q = A.B.(\overline{B} + C) + B.C + B$$
$$Q = A.B.\overline{B} + A.B.C + B.C + B$$
$$Q = A.0 + A.B.C + B.(C + 1)$$
$$Q = A.B.C + B.1$$
$$Q = A.B.C + B$$
$$Q = B.(A.C + 1)$$
$$Q = B.1$$
$$Q = B$$

5.

$$Q = B.(A + \overline{C}) + A + A.(\overline{A} + B)$$
$$Q = A.B + B.\overline{C} + A + A.\overline{A} + A.B$$
$$Q = A.B + B.\overline{C} + A$$
$$Q = A.(B + 1) + B.\overline{C}$$
$$Q = A + B.\overline{C}$$

6.

$$Q = A.B.\overline{C} + B.C + \overline{A}.B.\overline{C} + A.B.\overline{B}$$
$$Q = B.\overline{C}.(A + \overline{A}) + B.C + A.0$$
$$Q = B.\overline{C}.1 + B.C$$
$$Q = B.\overline{C} + B.C$$
$$Q = B.(\overline{C} + C)$$
$$Q = B.1$$
$$Q = B$$

7.

$$Q = A.B.C + B.C.D + B.C.\overline{D} + B.\overline{C}.D + A.B.\overline{C} + \overline{A}.B.\overline{C}$$
$$Q = A.B.C + B.C.(D + \overline{D}) + B.\overline{C}.D + B.\overline{C}.(A + \overline{A})$$
$$Q = A.B.C + B.C.1 + B.\overline{C}.D + B.\overline{C}.1$$
$$Q = A.B.C + B.C + B.\overline{C}.D + B.\overline{C}$$
$$Q = B.C.(A + 1) + B.\overline{C}.(D + 1)$$
$$Q = B.C.1 + B.\overline{C}.1$$
$$Q = B.C + B.\overline{C}$$
$$Q = B.(C + \overline{C})$$
$$Q = B.1$$
$$Q = B$$

8.

$$Q = A.B.C.D + A.B.D + A.\overline{B}.D + A.\overline{B}.C.D + A.C.D + \overline{A}.C.D$$
$$Q = A.B.D.(C + 1) + A.\overline{B}.D + A.C.D.(\overline{B} + 1) + \overline{A}.C.D$$
$$Q = A.B.D.1 + A.\overline{B}.D + A.C.D.1 + \overline{A}.C.D$$
$$Q = A.B.D + A.\overline{B}.D + A.C.D + \overline{A}.C.D$$
$$Q = A.D.(B + \overline{B}) + C.D.(A + \overline{A})$$
$$Q = A.D.1 + C.D.1$$
$$Q = A.D + C.D$$
$$Q = D.(A + C)$$

Exercise 5:

1.

$$Q = \overline{\overline{(A+\overline{B})}.\overline{(A.\overline{B})}} + \overline{A}.B$$

$$Q = \overline{\overline{(A+\overline{B})}} + \overline{\overline{(A.\overline{B})}} + \overline{A}.B$$

$$Q = (A+\overline{B}) + (A.\overline{B}) + \overline{A}.B$$

$$Q = A + \overline{B} + A.\overline{B} + \overline{A}.B$$

$$Q = A.(1+\overline{B}) + (\overline{B} + \overline{A}.B)$$

$$Q = A.1 + (\overline{B} + \overline{A})$$

$$Q = A + \overline{B} + \overline{A}$$

$$Q = \overline{B} + (A + \overline{A})$$

$$Q = \overline{B} + 1$$

$$Q = 1$$

2.

$$Q = \overline{\overline{\overline{A.\overline{C}}.\overline{B}}.D} + \overline{\overline{C}.D}$$

$$Q = \overline{\overline{\overline{\overline{A.\overline{C}}.\overline{B}}}} + \overline{D} + \overline{\overline{C}} + \overline{D}$$

$$Q = \overline{A.\overline{C}}.\overline{B} + \overline{D} + C$$

$$Q = (\overline{A} + \overline{\overline{C}}).\overline{B} + \overline{D} + C$$

$$Q = \overline{A}.\overline{B} + \overline{B}.C + \overline{D} + C$$

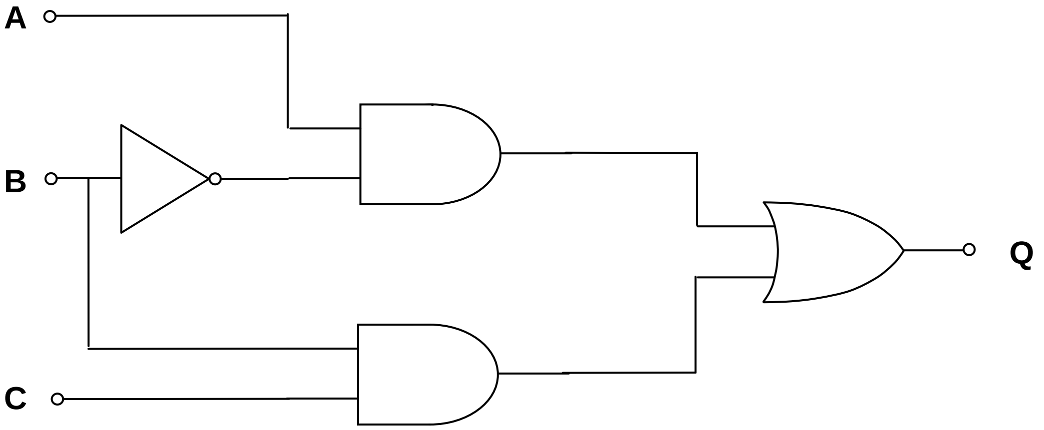
$$Q = \overline{A}.\overline{B} + C.(\overline{B} + 1) + \overline{D}$$

$$Q = \overline{A}.\overline{B} + C.1 + \overline{D}$$

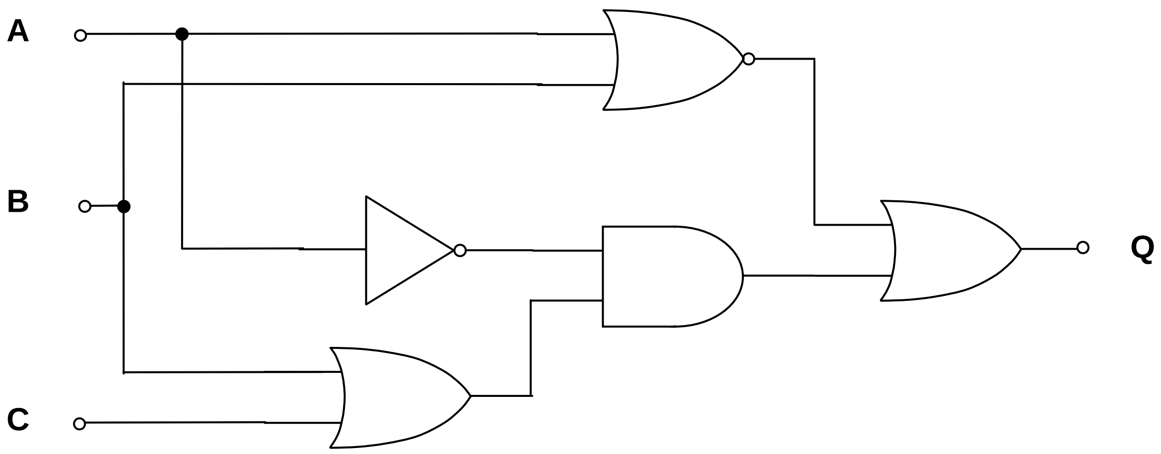
$$Q = \overline{A}.\overline{B} + C + \overline{D}$$

Exercise 6:

1. $Q = A.\overline{B} + B.C$



2. $Q = \overline{A + B} + \overline{A}.(C + B)$



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Examination Style Questions.

1. Simplify the following expressions, showing all stages of your working.

(a) $\mathbf{A} + \overline{\mathbf{A}} =$ ……………………………………………………………………………………

………

[1]

(b) $\mathbf{A}.\overline{\mathbf{B}} + \mathbf{B} =$ ……………………………………………………………………………………

………

[1]

(c) $\mathbf{A}.\mathbf{B} + \mathbf{B} =$ ……………………………………………………………………………………

………

[2]

(d) $\overline{\mathbf{A} + \overline{\mathbf{B}.\overline{\mathbf{C}}}} =$ ……………………………………………………………………………………

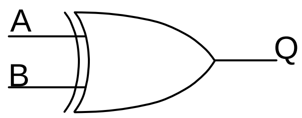
………

………

………

[2]

2. *(a)* Name, and complete the truth table for the logic gate whose symbol is shown below.

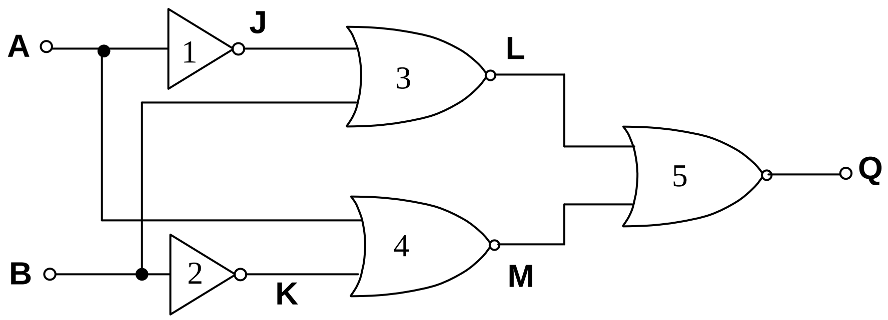


Name of gate : ……………………………………………….. [1]

Truth Table: [1]

A	B	Q
0	0	
0	1	
1	0	
1	1	

(b) (i) Complete the truth table for the following system.



[5]

A	B	J	K	L	M	Q
0	0					
0	1					
1	0					
1	1					

(ii) The system can be modified to perform the same function as the gate shown in part *(a)*. Explain what change needs to be made.

…………………………………………………………………………………………………

…………………………………………………………………………………………………

[1]

3. *(a)* Factorise the following Boolean expression, and simplify the result.

$$\mathbf{A.\overline{B} + \overline{A}.\overline{B}}$$

…………………………………………………………………………………………

…………………………………………………………………………………………

[2]

(b) Apply DeMorgan's theorem to the following expression, and simplify the result.

$$\overline{\mathbf{A.\overline{B}}}$$

…………………………………………………………………………………………

…………………………………………………………………………………………

[2]

(c) Use the results to *(a)* and *(b)* to show that

$$\mathbf{A.\overline{B} + \overline{A}.\overline{B} + \overline{A.\overline{B}} = \overline{A}}$$

…………………………………………………………………………………………

…………………………………………………………………………………………

…………………………………………………………………………………………

…………………………………………………………………………………………

[3]

4. Simplify the following expressions. Show all stages of your working for part *(c)* of the question.

(a) $\mathbf{A.0} =$ ……………………………………………………………………

[1]

(b) $\mathbf{A + A} =$ ……………………………………………………………………

[1]

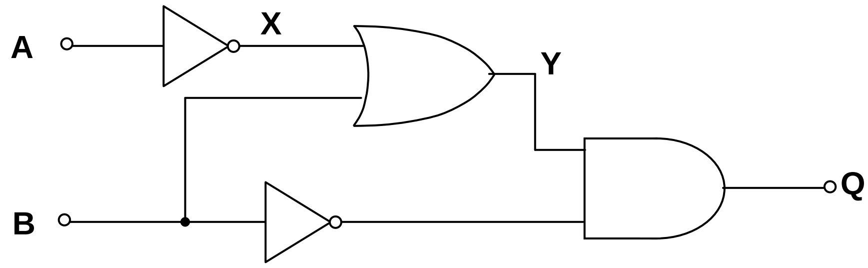
(c) $\mathbf{A.B + A} =$ ……………………………………………………………………

…………………………………………………………………………………………

…………………………………………………………………………………………

[2]

5. *(a)* Here is a system of logic gates:



Write down Boolean expressions for the outputs **X**, **Y** and **Q** in terms of the inputs **A** and **B**

X = ...

Y = ...

Q = ...

[3]

(b) Complete the following identities.

(i) $\mathbf{A.1} =$ ....

[1]

(ii) $\mathbf{A + 1} =$ ....

[1]

(iii) $\mathbf{A + \overline{A}} =$ ....

[1]

(c) Simplify the following expression using DeMorgan's theorem.

$$\overline{\mathbf{A.(\overline{B.A})}}$$

....

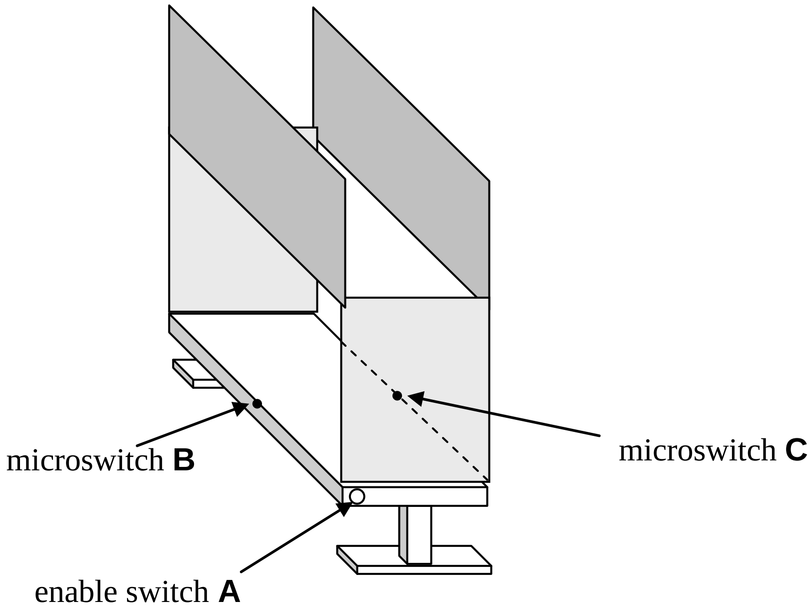
....

....

....

[2]

6. The diagram shows a cabinet designed to house machinery. Sliding panels allow easy access.



The output **Q** of an electronic warning system triggers an alarm if the enable switch **A** is turned on, when either panel is open, or both are open.

Each sliding panel closes a microswitch, when in place.
The microswitches, **B** and **C**, output a logic 0 signal when closed.
The enable switch **A** sends a logic 1 signal to the electronic system when turned on.
The system outputs a logic 1 to trigger the alarm.

(a) Complete the truth table for the logic system.

[2]

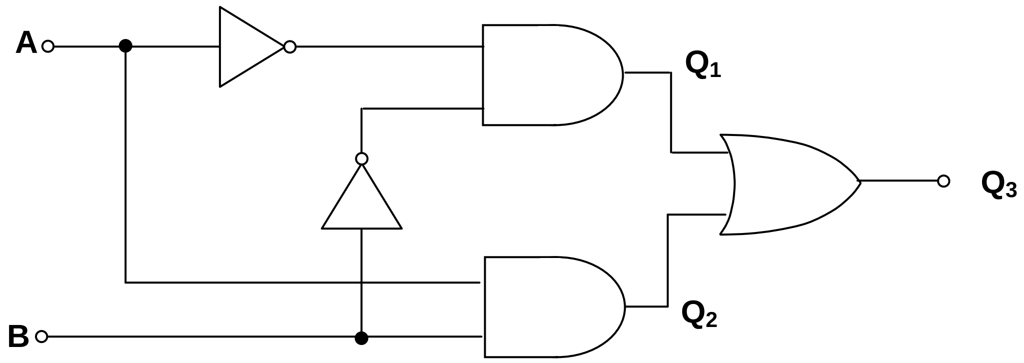
Microswitch (C)	Microswitch (B)	Enable Switch (A)	Output (Q)
0	0	0	
0	0	1	
0	1	0	
0	1	1	
1	0	0	
1	0	1	
1	1	0	
1	1	1	

(b) Write a Boolean expression for the output **Q** in terms of inputs **C**, **B** and **A**.
There is no need to simplify the expression

....

[2]

7. The following diagram shows a logic system.



(a) Give the Boolean expressions for **each** of the outputs **Q_1**, **Q_2** and **Q_3** in terms of the inputs **A** and **B.**

Q_1 = ...

Q_2 = ...

Q_3 = ...

[3]

(b) Complete the following truth table for **Q_1**, **Q_2** and **Q_3**.

[3]

INPUTS		OUTPUTS		
A	**B**	**Q_1**	**Q_2**	**Q_3**
0	**0**			
0	**1**			
0	**0**			
0	**1**			

(c) Name the single 2-input gate that could produce the same function as the **Q_3** output.

....

[1]

8. *(a)* Simplify the following expressions showing all appropriate working.

(i) $\mathbf{A.1} =$ ....

[1]

(ii) $\overline{\mathbf{B}}.(\mathbf{A}+\mathbf{B}) =$ ....

....

[2]

(b) Apply DeMorgan's theorem to the following expression and simplify the result.

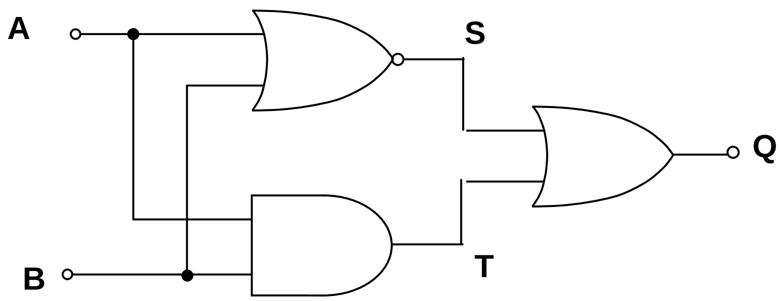
$(\overline{\mathbf{A}+\mathbf{B}})+\overline{\mathbf{A}} =$ ....

....

....

[3]

9. The diagram below shows a simple logic system.



(a) Complete the truth table for the signal at points **S**, **T** and **Q**.

B	A	S	T	Q
0	0			
0	1			
0	0			
0	1			

(b) (i) Name the single gate that can replace the combination of gates above.

....

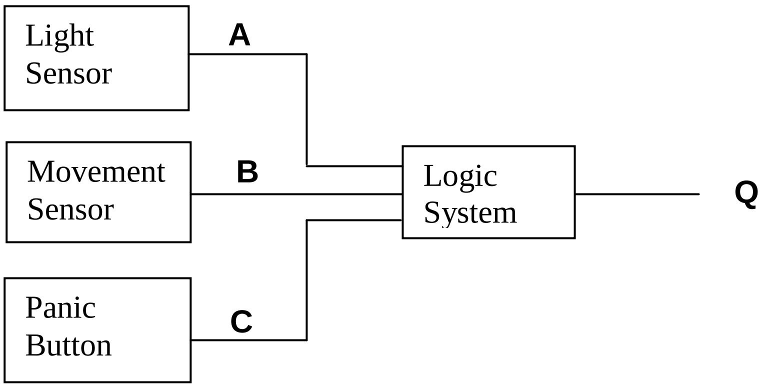
[1]

(ii) Draw the symbol for this gate.

[1]

10. (a) A security system is designed to switch on a powerful floodlight when movement is detected or a panic button is pressed. The system should operate **only** when it is dark. The specifications for the three input subsystems are given below:

Light Sensor	**A**	**Dark = Logic 0**
Movement Sensor	**B**	**Movement = Logic 1**
Panic Button	**C**	**Pressed = Logic 1**



(i) Complete the truth table for the logic system such that **Q = logic 1** when the floodlight is **on.**

[2]

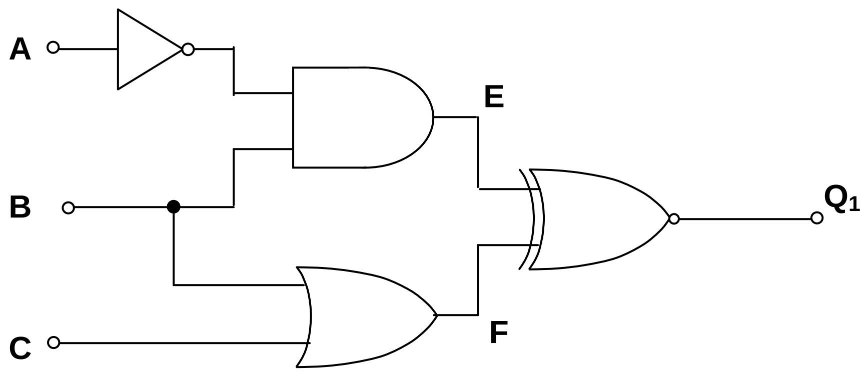
Panic Button (C)	Movement Sensor (B)	Light Sensor (A)	Output (Q)
0	0	0	
0	0	1	
0	1	0	
0	1	1	
1	0	0	
1	0	1	
1	1	0	
1	1	1	

(ii) Write down the Boolean expression for **Q**.

Q = ……………………………………………………………………………………

[2]

11. *(a)* A system of logic gates is shown below.



Give the Boolean expressions for the outputs **D**, **E** and **F** of the gates in the logic system **in terms of the inputs A and B.**

D = ...

E = ...

F = ...

[3]

(b) Complete the following truth table for **D**, **E**, **F**, and Q_1.

[4]

C	B	A	D	E	F	Q_1
0	**0**	**0**				
0	**0**	**1**				
0	**1**	**0**				
0	**1**	**1**				
1	**0**	**0**				
1	**0**	**1**				
1	**1**	**0**				
1	**1**	**1**				

(b) Use DeMorgan's theorem and Boolean Algebra to show that : $\overline{(\mathbf{A}+\mathbf{B}).(\mathbf{A}.\overline{\mathbf{B}})} = \overline{\mathbf{A}}+\mathbf{B}$

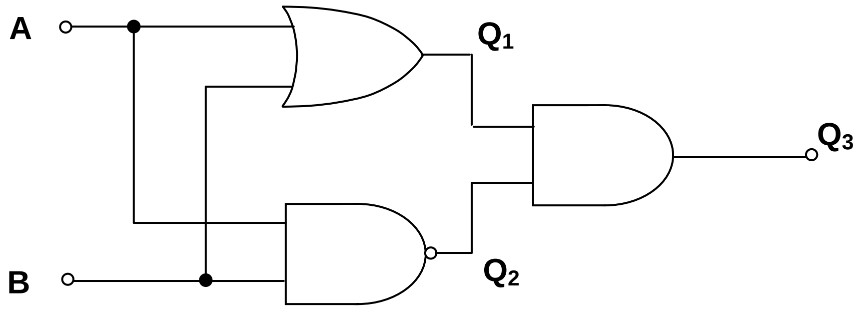
....

....

....

[3]

12. The following diagram shows a logic system.



(a) Write down the Boolean expressions for **each** of the outputs $\mathbf{Q_1}$, $\mathbf{Q_2}$ and $\mathbf{Q_3}$ in terms of the inputs **A** and **B.**

$\mathbf{Q_1}$ = ...

$\mathbf{Q_2}$ = ...

$\mathbf{Q_3}$ = ...

[3]

(b) Complete the following truth table for $\mathbf{Q_1}$, $\mathbf{Q_2}$ and $\mathbf{Q_3}$.

[3]

INPUTS		**OUTPUTS**		
A	**B**	$\mathbf{Q_1}$	$\mathbf{Q_2}$	$\mathbf{Q_3}$
0	**0**			
0	**1**			
0	**0**			
0	**1**			

(c) Name the single 2-input gate that could produce the same function as the $\mathbf{Q_3}$ output.

....

[1]

WJEC CBAC

13. A logic system must generate the output Q where

$$\mathbf{Q} = \overline{\mathbf{C}}.\overline{\mathbf{B}}.\mathbf{A} + \overline{\mathbf{B}}.\overline{\mathbf{A}}$$

Complete the truth table for this system.

[2]

C	B	A	Q
0	0	0	
0	0	1	
0	1	0	
0	1	1	
1	0	0	
1	0	1	
1	1	0	
1	1	1	

14. *(a)* Simplify the following expressions showing all appropriate working.

(i) $\mathbf{A}.\mathbf{0} =$ ....

[1]

(ii) $\mathbf{A} + \overline{\mathbf{A}}.\mathbf{B} =$ ....

....

[1]

(b) Apply DeMorgan's theorem to the following expression **and** simplify the result.

$\overline{\overline{\mathbf{A}} + (\overline{\mathbf{B}.\mathbf{A}})} =$ ....

....

....

....

....

[3]

Module ET1
Introduction to Analogue and Digital Systems.

Self Evaluation Review

Learning Objectives	My personal review of these objectives: ☺	😐	☹
Generate the Boolean expression for a system [with up to 3 inputs] from a truth table;			
Generate the Boolean expression for a system [with up to 4 inputs] from a logic diagram;			
Recall and use the following identities ○ $\mathbf{A.1 = A}$ ○ $\mathbf{A.0 = 0}$ ○ $\mathbf{A.\overline{A} = 0}$ ○ $\mathbf{A + 1 = 1}$ ○ $\mathbf{A + 0 = A}$ ○ $\mathbf{A + \overline{A} = 1}$			
Select and use the following identities ○ $\mathbf{A + \overline{A}.B = A + B}$ ○ $\mathbf{A.B + A = A(B + 1) = A}$			
Apply DeMorgan's theorem to simplify a logic system [with up to 3 inputs]			

Targets: 1. ....

....

2. ....

....